Optimizing Price Prediction: A Model Selection Approach to Tucson Housing Prices

Abstract

This project develops a predictive model to determine the most significant drivers of housing prices in Tucson, Arizona. Using public sales and property characteristic data for the 2025 tax year from the Pima County Assessor's Office, a multiple linear regression model was constructed. A backward stepwise selection method, based on the Akaike Information Criterion (AIC), was employed to identify the optimal combination of predictors. The final model explains 33.9% of the variance in sale prices and identifies square footage, year built, number of stories, number of rooms, number of bathroom fixtures, number of patios, pool area, and property type as the most statistically significant predictors. The findings provide a robust, data-driven framework for property valuation in this market.

Introduction

The valuation of residential property is a complex process influenced by numerous characteristics. This project seeks to demystify this process for the Tucson, Arizona housing market by creating an optimized predictive model. The primary research question is: "What is the optimal combination of property characteristics that creates the most accurate and parsimonious model for predicting housing sale prices in Tucson, AZ?"

The data for this analysis was sourced entirely from the Pima County Assessor's Office public data portal (https://www.asr.pima.gov/downloads-data), a government-maintained repository of property information. Two primary datasets from the 2025 tax year were utilized: Sale2025.csv, containing property sales transactions, and Mas25.csv, containing detailed residential property characteristics.

Methodology

The analysis was performed in the R programming language using RStudio, with the tidyverse package for data manipulation and visualization. The Sale2025.csv and Mas25.csv datasets were loaded into R. They were then merged into a single comprehensive dataframe using their common identifier—the property parcel number. The merged dataset was cleaned to ensure data quality for modeling. This involved filtering out invalid sales and properties with unrealistic building areas. The categorical variable SFRCONDO was converted into a factor data type and the continuous variables SalePrice, SQFT, YEAR, STORIES, ROOMS, BATHFIXTUR, PATIONUMBE, GARAGECAPA, and POOLAREA were converted into numeric types. To find the best predictive model, a backward stepwise regression procedure was used. This method begins with a "full" model containing all candidate predictors, and then iteratively removes the least statistically significant predictor at each step. The selection criterion used to guide this process was the Akaike Information Criterion (AIC).

Figure 1. Data Dictionary

Variable Name	Data Type	Description
SalePrice	Numeric	The final price the property sold for, in U.S. dollars.
SQFT	Numeric	The total finished living area of the house in square feet.
YEAR	Numeric	The year the property was originally built.
STORIES	Numeric	The number of stories or floors in the house.
ROOMS	Numeric	The total number of rooms in the house.
BATHFIXTUR	Numeric	The total count of bathroom fixtures (e.g., sink, toilet, shower).
PATIONUMBE	Numeric	The number of patios on the property.
GARAGECAPA	Numeric	The garage's capacity, measured by the number of cars it can hold.
POOLAREA	Numeric	The area of the swimming pool in square feet (0 indicates no pool).
SFRCONDO	Categorical	The type of property (e.g., Single-Family Home (S) or Condo (C)).

Results

Initial exploratory visualizations revealed key characteristics of the data. A histogram of SalePrice showed a strong right-skewed distribution, which was addressed for visualization by capping extreme values for homes priced at one million dollars or more.

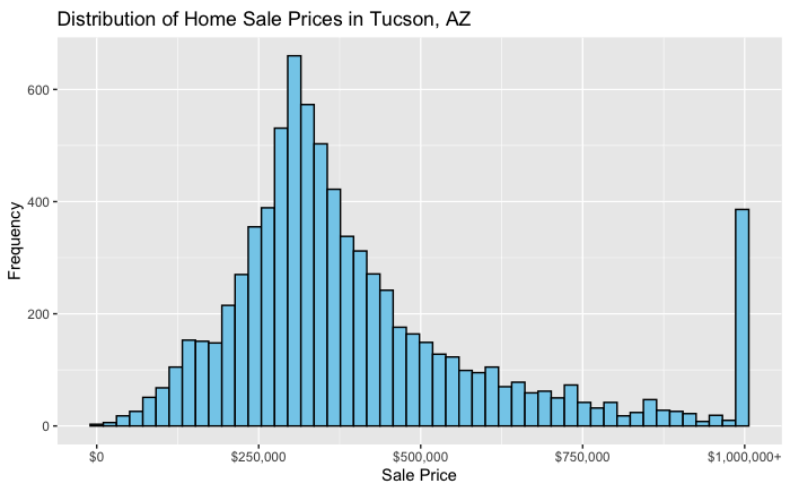

Figure 2. Histogram of the Distribution of Home Sale Prices in Tucson, AZ

The summary of the best fitting linear model after the selection process is presented below:

```
Call:
lm(formula = SalePrice ~ SQFT + YEAR + STORIES + ROOMS + BATHFIXTUR +
    PATIONUMBE + POOLAREA + SFRCONDO, data = model_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1201347   -98666   -26322    46196  5338156

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.969e+06  5.101e+05   5.820 6.13e-09 ***
SQFT         2.632e+02  9.273e+00  28.380  < 2e-16 ***
YEAR        -1.131e+03  2.014e+02  -5.615 2.03e-08 ***
STORIES     -1.806e+05  1.375e+04 -13.140  < 2e-16 ***
ROOMS       -6.027e+02  1.846e+02  -3.266 0.001096 **
BATHFIXTUR   1.890e+04  2.644e+03   7.146 9.70e-13 ***
PATIONUMBE   1.776e+04  5.148e+03   3.450 0.000564 ***
POOLAREA     1.159e+02  2.293e+01   5.054 4.41e-07 ***
SFRCONDOC   -1.247e+05  3.205e+05  -0.389 0.697151
SFRCONDOS   -7.153e+05  3.235e+05  -2.211 0.027031 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 320200 on 7935 degrees of freedom
Multiple R-squared:  0.3399,    Adjusted R-squared:  0.3392
F-statistic:   454 on 9 and 7935 DF,  p-value: < 2.2e-16
```

Figure 3. Summary output of the best fitting linear model after selection

The Adjusted R-squared is 0.3392, indicating that this model explains 33.9% of the variability in home sale prices. While this is lower than models that include the actual assessed property value, it represents a more authentic measure of the predictive power of the property's physical characteristics alone. The stepwise process resulted in a model where nearly every predictor is statistically significant at the 5% significance level. The final model was selected using a backward stepwise procedure based on the Akaike Information Criterion (AIC) to optimize predictive accuracy. The final model's regression equation can be written as:

$$\widehat{SalePrice} = 2,969,000 + 263.2(SQFT) - 1,131(YEAR) - 180,600(STORIES) - 602.7(ROOMS) + 18,900(BATHFIXTUR) + 17,760(PATIONUMBE) + 115.9(POOLAREA) - 124,700(I_{Condo}) - 715,300(I_{SFR})$$

After identifying the optimal model through stepwise selection, an ANOVA test was performed to compare it against the full model with all candidate predictors:

```
Analysis of Variance Table

Model 1: SalePrice ~ SQFT + YEAR + STORIES + ROOMS + BATHFIXTUR + PATIONUMBE +
    GARAGECAPA + POOLAREA + SFRCONDO
Model 2: SalePrice ~ SQFT + YEAR + STORIES + ROOMS + BATHFIXTUR + PATIONUMBE +
    POOLAREA + SFRCONDO
  Res.Df        RSS Df   Sum of Sq      F Pr(>F)
1   7934 8.1357e+14
2   7935 8.1357e+14 -1 -2239179465 0.0218 0.8825
```

Figure 4. Summary output of ANOVA test comparing the full model with the best model

The results (p-value = 0.8825) showed no statistically significant difference between the two models, confirming that the variable GARAGECAPA removed by the stepwise procedure was not contributing significant predictive power. Therefore, the more parsimonious model was selected for the final analysis, as it provides the same explanatory power with fewer variables.

Conclusion

This analysis successfully identified an optimal, methodologically sound model for predicting housing prices in Tucson, AZ. The research question was answered by determining that the best combination of predictors includes a property's square footage, its age and type, the number of stories, rooms, bathroom fixtures, patios, and pool area. The findings highlight the importance of proper variable selection to avoid circular reasoning and build a truly explanatory model. While the model's predictive power is moderate, it provides a valid foundation for understanding the housing market. Future research could improve accuracy by incorporating location data, like property zip code, or exploring non-linear relationships and interaction terms between predictors. For example, whether the impact of certain features, like square footage, is consistent across different price points can be explored using a more advanced model that incorporates an interaction term.

References

Pima County Assessor's Office. (2025). *Public Records Data*. Retrieved from https://www.asr.pima.gov/downloads-data

R Core Team (2025). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/.

Wickham et al., (2019). Welcome to the tidyverse. *Journal of Open Source Software, 4*(43), 1686. https://doi.org/10.21105/joss.01686

Google Gemini. (2025). AI assistance for data analysis, code troubleshooting, and report generation.

Appendix

Complete R script from the Midterm Project.Rmd file:

Data Analysis and Preparation

```
# 1. Load Libraries
library(tidyverse)

# 2. Load and Merge Datasets
sales_data <- read.csv("Sale2025.csv")
res_data <- read.csv("Mas25.csv")

full_data <- inner_join(sales_data, res_data, by = c("Parcel" = "PARCEL"))
head(full_data)

# 3. Clean and Prepare Data
# create a list of all column names that should be numbers
numeric_columns <- c("SalePrice", "SQFT", "YEAR", "STORIES", "ROOMS",
"BATHFIXTUR", "PATIONUMBE", "GARAGECAPA", "POOLAREA")

model_data <- full_data %>%
  # convert data types for all necessary columns
  mutate(
    across(all_of(numeric_columns), as.numeric),
    SFRCONDO = as.factor(trimws(SFRCONDO))
    ) %>%
  # filter to keep SalePrice greater than $1000 and SQFT greater than 100 sqft
  filter(
    SalePrice > 1000,
    SQFT > 100
  ) %>%
  # keep only columns needed for the model
  select(all_of(numeric_columns), SFRCONDO) %>%
  # remove rows with remaining missing values
  na.omit()
head(model_data)

# 4. Exploratory Data Visualization
model_data %>%
  mutate(SalePrice_capped = ifelse(SalePrice > 1000000, 1000000, SalePrice)) %>%

  ggplot(aes(x = SalePrice_capped)) +
  geom_histogram(bins = 50, fill = "skyblue", color = "black") +

  scale_x_continuous(
    breaks = seq(0, 1000000, by = 250000),
```

```
  labels = c("$0", "$250,000", "$500,000", "$750,000", "$1,000,000+")) +

 labs(title = "Distribution of Home Sale Prices in Tucson, AZ",
      x = "Sale Price",
      y = "Frequency")
```

Stepwise Model Selection

```
# fit the full model with all candidate predictors
full_model <- lm(SalePrice ~ ., data = model_data)
require(faraway)
sumary(full_model)

# perform backward stepwise selection and find one with the lowest AIC
best_model <- step(full_model, direction = "backward", trace = 0)
summary(best_model)

# perform ANOVA test to see if the two models are significantly different
anova(full_model, best_model)
```